

January 2026 CSE 220
Online on Signals and Their Properties
Subsections:
Total Marks: 10
Total Time: 30 Minutes

In this assignment, you need to implement time reversal and amplitude scaling of a sampled continuous-time signal in Python. You also need to plot the graphs of each of these signals.

Let the base signal be $x(t)$ (provided in the template file). You will compute and plot:

$$x(t), y(t) = \alpha x(-t)$$

Tasks

- ★ Generate the time axis t and compute $x(t)$. (given)
- ★ Implement a function *transform_signal(t, x, alpha)* that produces $y(t) = \alpha x(-t)$
- ★ Assume the signal is bounded between $-\pi$ and π , values outside the range are equal to 0.
 - In case of time reversal, the signal will be reversed by the y-axis. For example: $y(0) = x(0)$, $y(1) = x(-1)$, $y(-1) = x(1)$ and so on.
 - In case of amplitude scaling, the signal's amplitude is compressed or expanded. For example: for $\alpha = 2$; $y(0) = 2x(0)$, $y(1) = 2x(1)$ and so on.
- ★ Plot (with proper labels and legend) on the same figure: $x(t)$, $y(t)$.
- ★ In the main function, make an infinite loop that will take alpha repeatedly, and show plots on each iteration. The program will end upon receiving 'q'.

Marks Distribution

- Implementing reversal: 4
- Implementing scaling: 4
- Completing the loop: 2